(Mobile Development w/Lab)
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Community Learning Center Management System

Short Description

A Dart console application that manages a community learning center. It registers learning resources (books, courses, workshops, equipment) and participants, processes borrowing and return transactions with membership discounts, and displays organized records and a summary report.

Implemented Features

- Menu-driven interface with 10 options, running in a loop until the user chooses Exit.
- Add learning resource – title, category, number of copies, daily rental rate.
- Register participant – name and membership (Regular or Premium, 20% discount).
- Borrow resource – fee is computed as rate × days, minus the membership discount. A participant may hold at most 3 active loans, and resources with no copies left cannot be borrowed.
- Return resource – restores stock and the participant's loan allowance.
- View resources, participants, transactions, or all records together.
- Summary report – totals, active loans, revenue, and resources per category.
- Input validation and error handling – invalid menu choices, empty text, non-numeric values, out-of-range numbers, unknown IDs, and closed input are handled with messages; the program never crashes (try/on FormatException, outer try/catch).

Object-Oriented Design

Concept	Where it is used
Abstract class	CenterRecord (abstract getSummary() and recordType)
Inheritance / overriding	LearningResource, Participant, Transaction each extend CenterRecord and implement getSummary()
Polymorphism	displayRecords() accepts a List<CenterRecord> and each object prints itself
Mixin	Timestamped gives every record a creation time
Encapsulation	Private fields (_title, _activeLoans, ...) with getters and validated setters
Constructors	Every class has a constructor that also generates its unique ID (R001, P001, T001)
List and Map	Map<String, LearningResource> for resources; List<Participant> and List<Transaction>; a Map counts resources per category
Enum	Membership (Regular / Premium) with its discount rate
User-defined functions	computeFee(), formatMoney(), displayRecords(), readInt(), readDouble(), readText(), and menu action functions

Sample Test Scenarios

Scenario 1 – Normal flow (add, register, borrow)

1. Choose 1 → Title Dart Basics, category 1 (Book), copies 2, rate 20.
2. Choose 2 → Name Ana Cruz, membership 2 (Premium).
3. Choose 3 → participant P001, resource R001, days 5. Expected: base fee PHP 100.00, discount 20%, amount due PHP 80.00; available copies become 1/2.

Scenario 2 – Return and summary

1. Choose 4 → transaction T001. Expected: status changes to *Returned*; copies return to 2/2.
2. Choose 9. Expected: 1 resource, 1 participant, 1 transaction (0 active), revenue PHP 80.00, Book: 1.

Scenario 3 – Invalid input and error conditions

- Menu choice abc or 15 → "Invalid choice" message; menu is shown again.
- Add resource with copies abc → "Invalid input. Please enter a whole number."; copies 0 → "Please enter a number from 1 to 100."
- Rate -5 → "Please enter a valid amount of 0 or more."
- Empty title → "Input cannot be empty."
- Borrow with participant ID P999 → "Participant not found. Transaction cancelled."
- Borrow when no copies remain, or a 4th active loan → clear refusal message.
- Choose 3 before any records exist → a message explaining a participant and resource are needed (try this by emptying demoInputs and starting with '3').

How to Run

DartPad (dartpad.dev)

1. Open dartpad.dev and start a new Dart project.
2. Replace all code with the contents of main.dart and click Run.
3. DartPad cannot read the keyboard, so main.dart feeds the three test scenarios above through the demoInputs list (see "Console Input Helpers" in the code). All prompts,

validation, calculations and output come from the program logic; edit demoInputs to try other inputs.

Interactive keyboard input (Dart SDK)

1. Install the Dart SDK.
2. Use the same program with real keyboard input: dart run main_stdin.dart (identical logic; only the input helper reads from stdin).

Requires Dart 2.17 or newer (uses an enhanced enum).